

ERES-Projekt

Plant Health System

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Project Plant Health System – Functional Description

- Connect the ADC Click Board to the Discovery Kit via the mikroBUS interface.
- Attach a soil moisture sensor to the ADC Click Board.
- Provide ADC measurement data from the Fabric to the MSS
- Process the digital data in the MSS.
- Store and Read physical measurement values in the Fabric SRAM through the MSS.
- Visualize the measurement data in Python via a UART interface.
- PCB Design: Reverse-engineer the ADC Click Board and modify the circuit design to support differential temperature measurements. The mechanical connector layout was also adapted to meet the new requirements.



Design Specification: Requirements and Verification

Requirement Section				Verification Section	
ID	Requirement	Source	Compliance	Type	Comment
1	The system shall provide one channels for soil moisture measurement capable of recording values across a range from 0% to 100%.	User Requirement	FC	R+T	Review of Software, Tested with Serial Terminal and Python
2	The soil moisture measurement shall achieve a resolution of at least 1%.	User Requirement	FC	R+T	Review of Software, Tested with Serial Terminal and Python
3	The system shall provide one channel for temperature measurement capable of recording values across a range of at least 0 °C to 70 °C.	User Requirement	FC	A	Calculation for circuit design
4	The temperature measurement shall achieve a resolution of at least 0.5 °C.	User Requirement	FC	R	Review of component datasheets for the temperature measurement circuit
5	The system shall provide at least two additional, unassigned connection interfaces to support the integration of future sensors (e.g., light intensity measurement).	User Requirement	FC	R	Review of ADC-Datasheet
6	Soil moisture measurement should be performed every 1 ms.	User Requirement	FC	T	SPI Analysis: Measuring time between Measurements
7	The system shall update and store the measured sensor values at a fixed interval of every 10 seconds (+/-1s tolerance)	User Requirement	FC	T	Counting Time and operate with Serial Terminal or Python
8	The system shall continuously display sensor values in physical units over a time chart for a minimum duration of one hour after power up	User Requirement	FC	T	Testing with Python
9	The system is permitted to use volatile memory for storing the measurement data.	User Requirement	FC	R	Review of Implementation
10	The system shall use the PolarFire SoC Discovery Kit as the primary control unit.	User Requirement	FC	R	Review of Implementation
11	The physical and logical connection between the ADC and the PolarFire SoC Discovery Kit shall be implemented via the mikroBUS interface.	System Requirement	FC	R	Review of Implementation
12	The system shall provide a UART-based data transfer interface to establish a connection between the PolarFire SoC Discovery Kit and a PC.	System Requirement	FC	R	Review of Implementation
13	The system shall provide an Analog-to-Digital Converter (ADC) that meets the following minimum criteria: At least 4 available channels, A resolution of at least 8 bits, A sampling rate of at least 1000 S/s (Sample per second) per channel, operater with a voltagelevel (analog input) 0...3.3V	System Requirement	FC	R + T	Review Datasheet of ADC, Test voltage range with Potentiometer, Multimeter and Debug-Session
14	The system shall provide a storage with a minimum memory depth of 4096 words and a data width of at least 32 bits to store measurement data for a minimum interval of 10 seconds and duration of one hour per channel. If only one channel is used, a depth of 1024 may be configured.	System Requirement	FC	A	Calculation of memory usage

Design Specification: Interface Specification

Interfaces Specification						
Component	Component-Pin	Function	MPFS Pin	Bank	Logic/SW Name	Description
MCP3204-CI/SL	CS/SHDN	Communication for issuing commands for ADC conversion and reading out results	C16	Bank 1	Mikrobus_Cs	MCP3204 ADC SPI Interface to FPGA.
MCP3204-CI/SL	DOUT		A18	Bank 1	Mikrobus_Miso	
MCP3204-CI/SL	DIN		E11	Bank 1	Mikrobus_Mosi	
MCP3204-CI/SL	CLK		E17	Bank 1	Mirkobus_Sck	

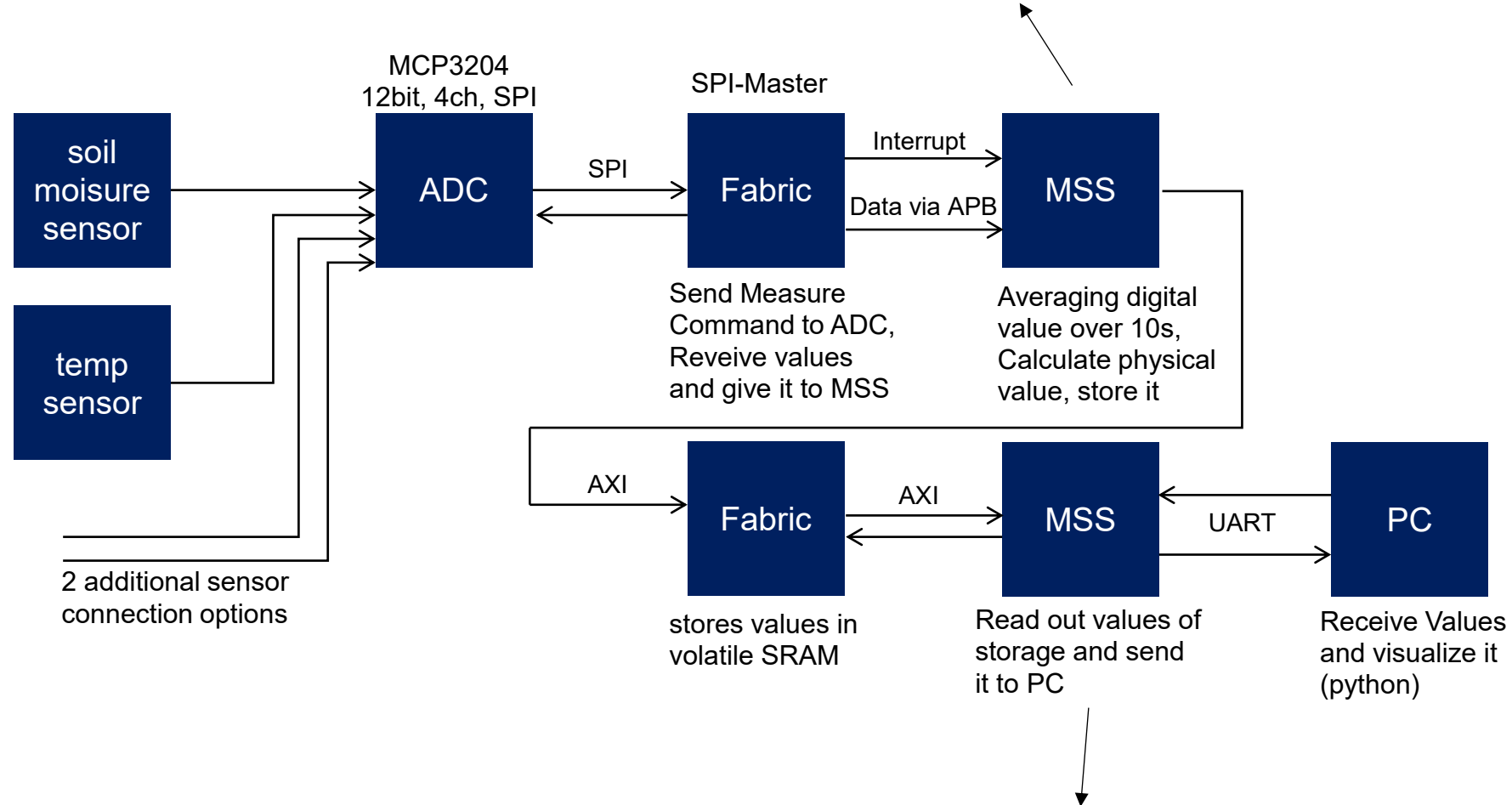
Trade-off between capacitive and resistive soil moisture sensors

Criteria	Item Weight	Group Weight / Norm.	Rating (SparkFun)	rate x weight (SparkFun)	Flags Risk (SparkFun)	Rating (AZDelivery)	rate x weight (AZDelivery)	Flags Risk (AZDelivery)	Remarks
Performance & Hardware		3							
Corrosion resistance	3	1,80	-1	-2	X	1	2		Risk: SparkFun is resistive and corrodes over time in moist soil
Signal stability	2	1,20	1	1		-1	-1	X	Risk: Cheap capacitive clones often have faulty voltage regulators
Integration & Usability		3							
Mechanical stability	2	2,00	1	1		-1	-2		SparkFun features a thick and rugged PCB
Connectivity	1	1,00	0	0		0	0		Both come with bare pins by default (soldering required)
Development Effort		2							
Testability	2	2,00	3	2		1	2		A resistive sensor can be tested using a potentiometer
Programmatic		1							
Costs	3	1,00	1	1		2	2		AZDelivery is cheaper in packs
Result				2,9			2,6		

Design Description

Core0: ISR: Read APB, store digital Data in variable, increment counter

Core1: Detects a counter increment, accumulates 10000 digital data samples, averaging, calculates the physical value, and stores the result in SRAM via AXI.



SRAM

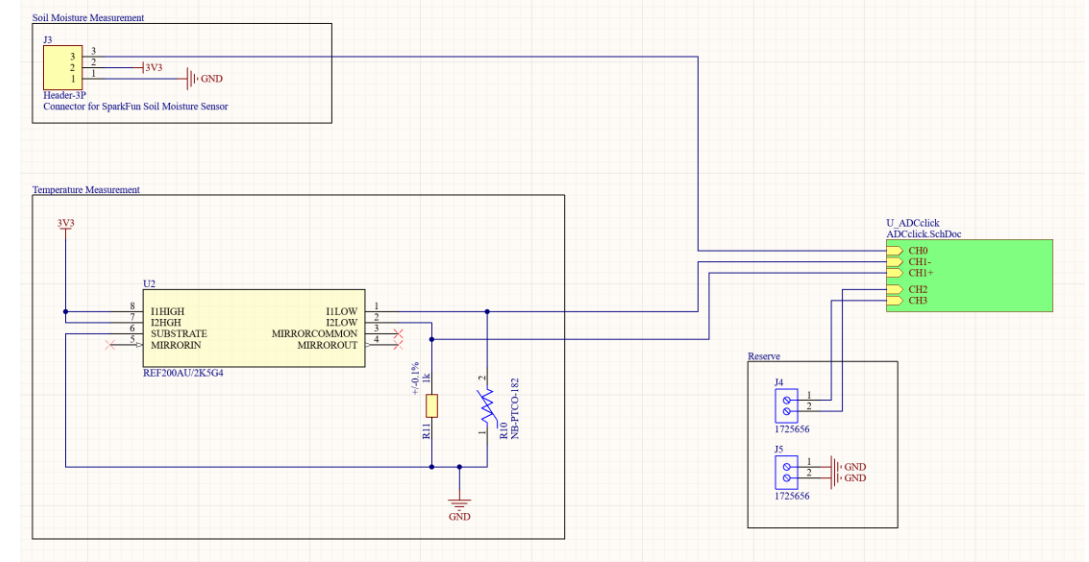
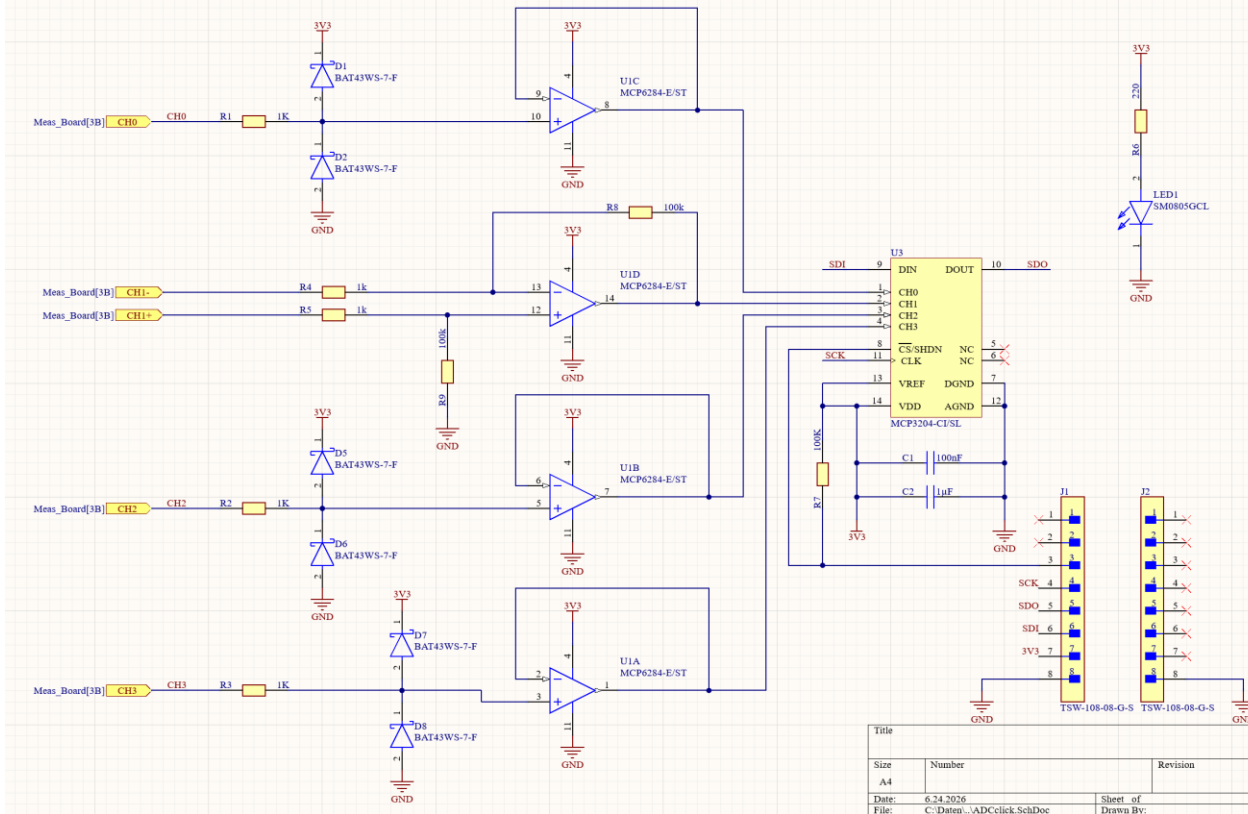
Index	Value 1
Value 2	Value 3
Value 4	
	⋮

Core3: Receives user commands (COUNT, DUMP) via UART0 and reads data from SRAM.

COUNT: Returns the number of stored measurement values.

DUMP: Exports all stored measurement values in CSV format.

Adaptation in the circuit diagram (Review Requirement 3)

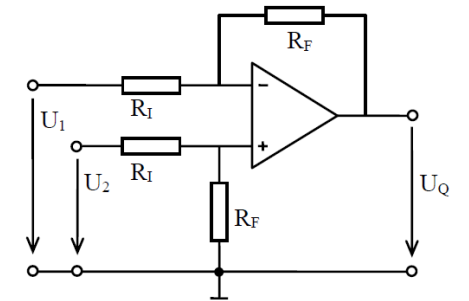


Normalerweise wählt man $R_1 = R_2 = R_I$ und $R_3 = R_4 = R_F$

Dann gilt:

$$U_Q \approx (U_2 - U_1) \cdot \frac{R_F}{R_I}$$

Abbildung 15-13
Differenzverstärker



- PT1000 mit 3850 ppm/°C → 0°C = 1000 Ω, Changes: 0.3851Ω / 1°C
- Current-Source 100µA at each Channel
- Measuring diff between constant 1000Ω and PT1000
- The amplifier has a gain factor of 100
- For Temperatures 0...70°C the analog-voltage (after amplifier) is 0 ... 2.695V

SRAM: (Review Requirement 14)

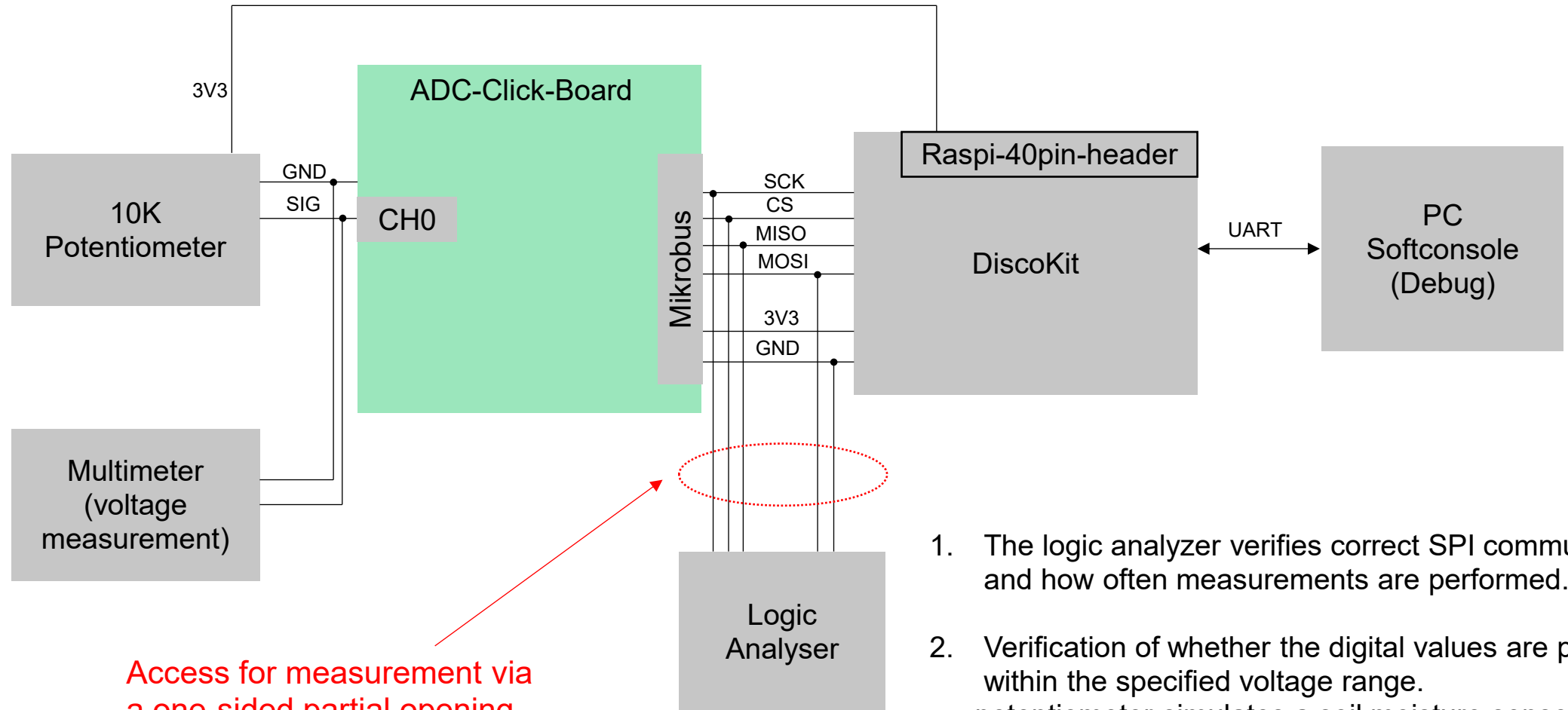
1. Configurations at 64-Bit Data Width

Configuration	Memory Depth	Memory Capacity [KiB]	Max. Datensätze	Max. Runtime (at 10s interval) [s]	Max. Runtime (at 10s interval)
Test (Actual)	1.024	8	2047	20470	5h 41m 10s
Short	2.048	16	4.095	40.950	11h 22m 30s
Medium	4.096	32	8.191	81.910	22h 45m 10s
Extended	8.192	64	16.383	163.830	1d 21h 30m 30s
Longterm	16.384	128	32.767	327.670	3d 19h 01m 10s

2. Memory Map at Memory Depth 1.024

Word-Index (dez)	Byte Adress (hex)	Payload
0	0x0000_0000	INDEX und Dataset #1
1	0x0000_0008	Dataset #2 und #3
2	0x0000_0010	Dataset #4 und #5
3	0x0000_0018	Dataset #6 und #7
... Fortlaufend ...		
1022	0x0000_1FF0	Dataset #2044 und #2045
1023	0x0000_1FF8	Dataset #2046 und #2047

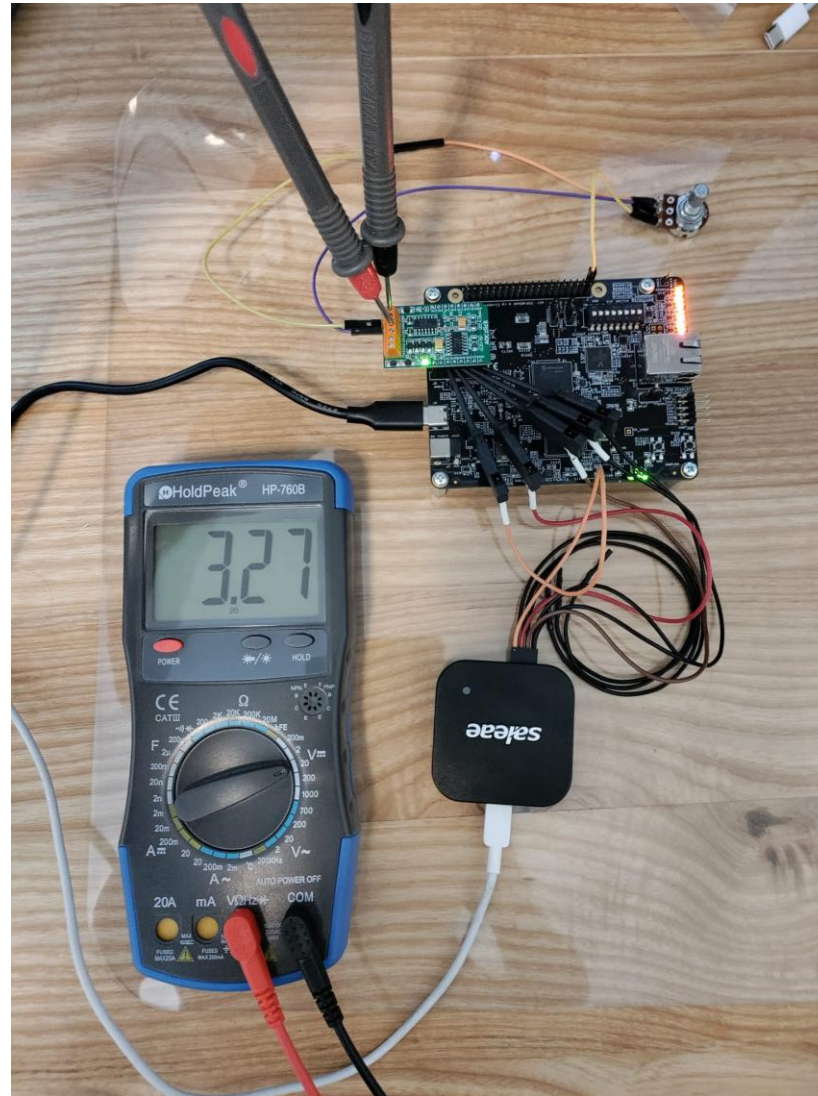
Verification Plan (Requirement 6 & 13)



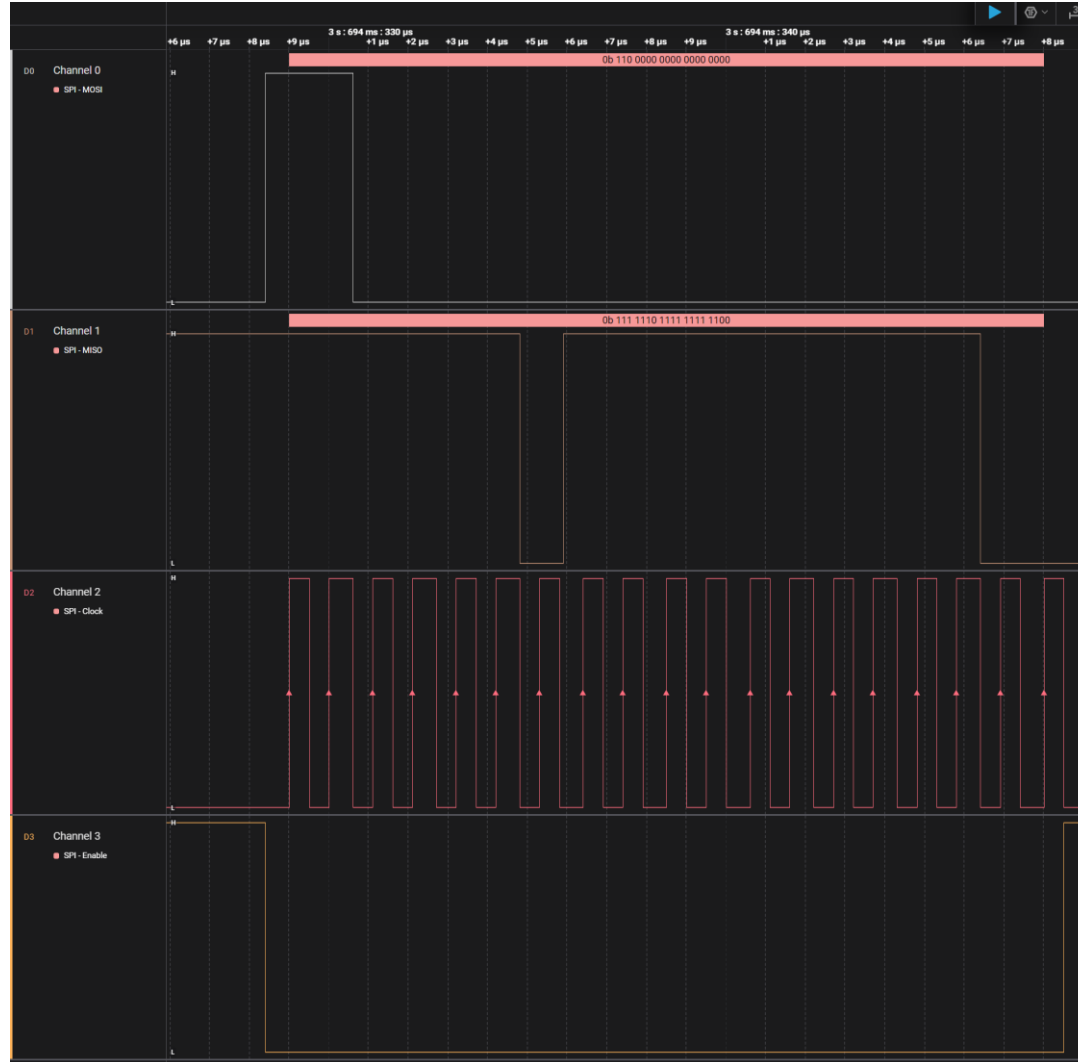
Access for measurement via a one-sided partial opening, while maintaining reliable contact.

1. The logic analyzer verifies correct SPI communication and how often measurements are performed.
2. Verification of whether the digital values are plausible within the specified voltage range.
 - potentiometer simulates a soil moisture sensor
 - multimeter measures the voltage at CH0 relative to GND
 - PC running a software console in debug mode receives the digital values

Verification Plan (Requirement 6 & 13) as Picture



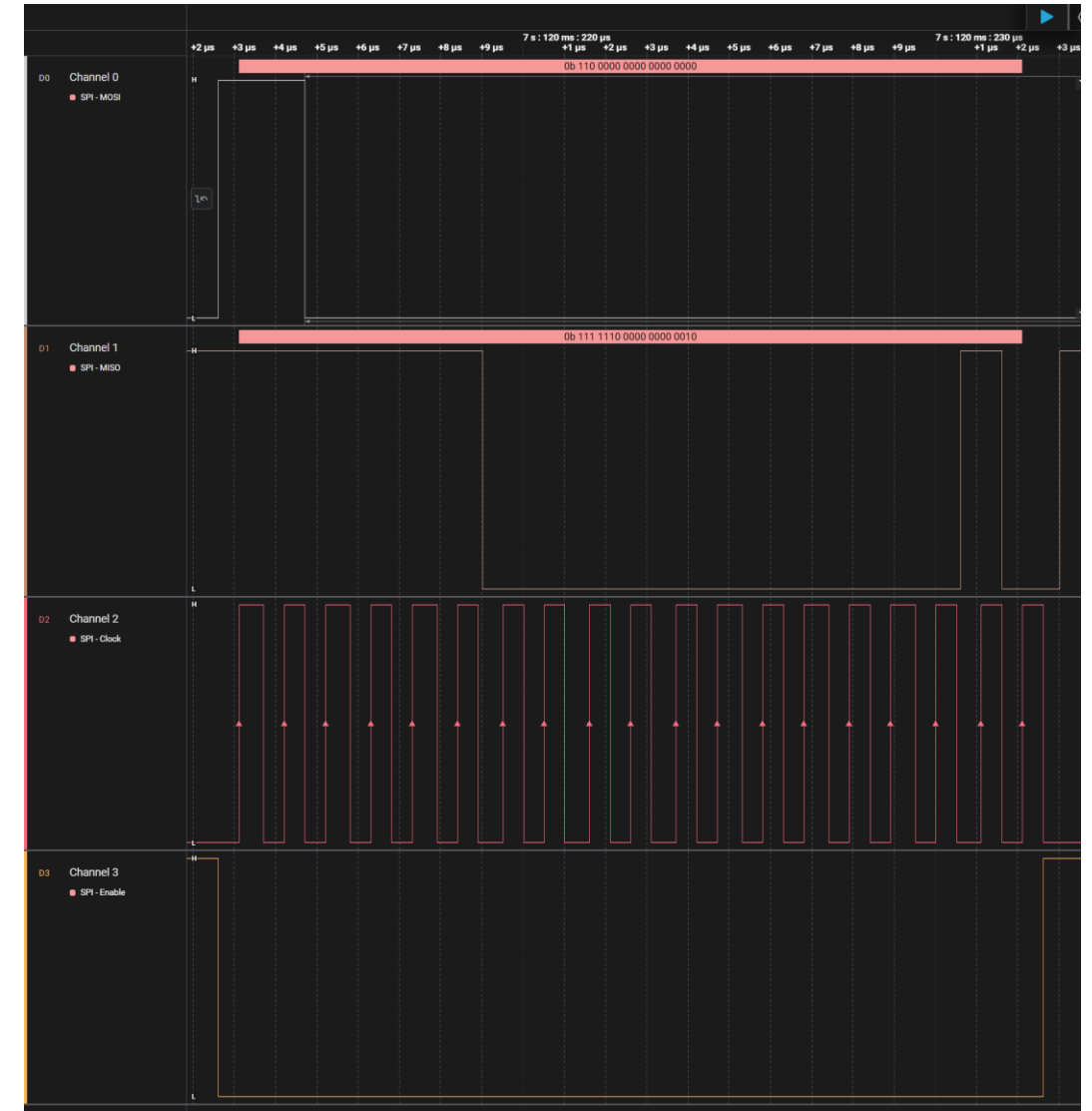
Verification Results: SPI Analysis



$$U_{CH0} = 3.27V$$



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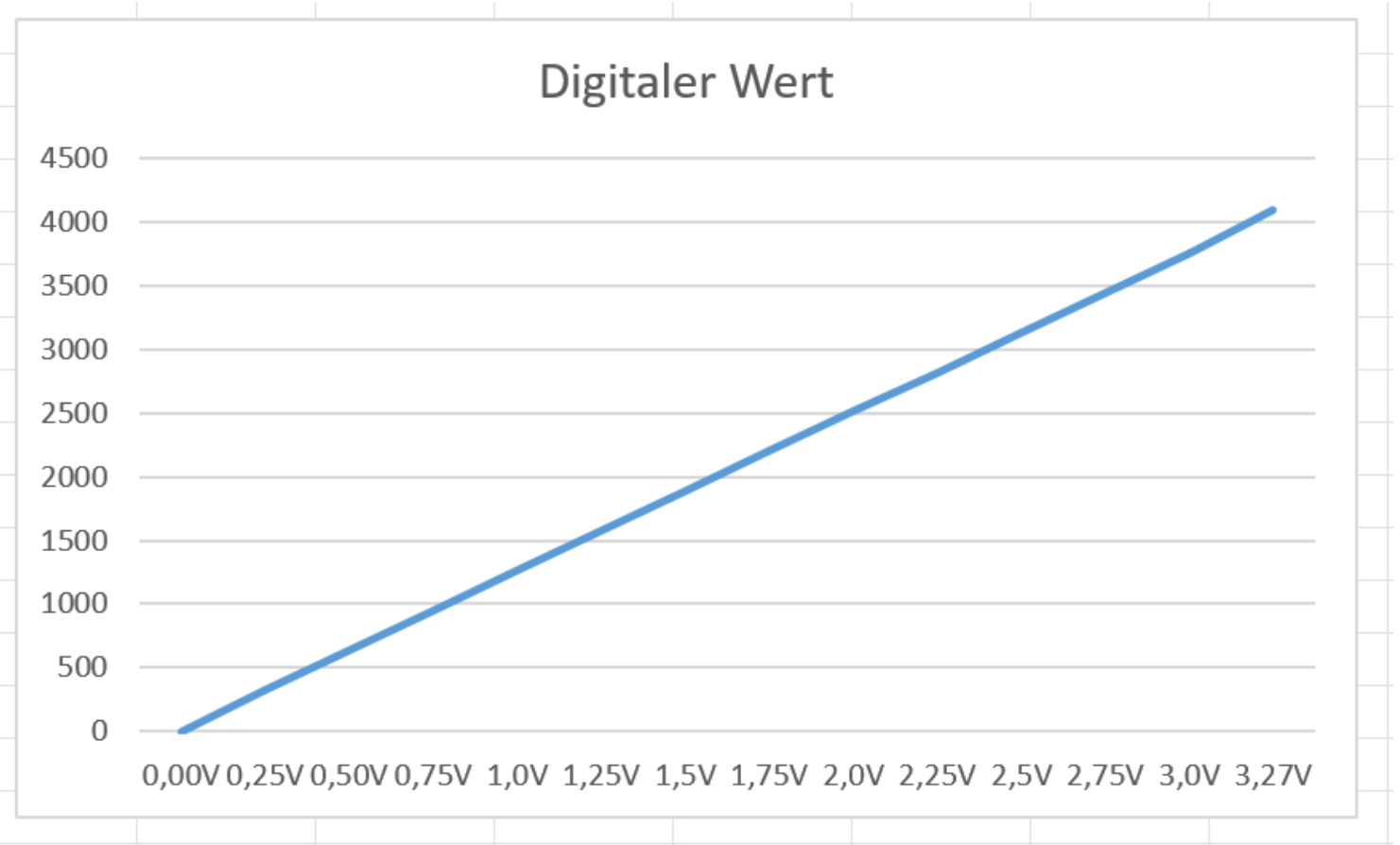


$$U_{CH0} = 0V$$

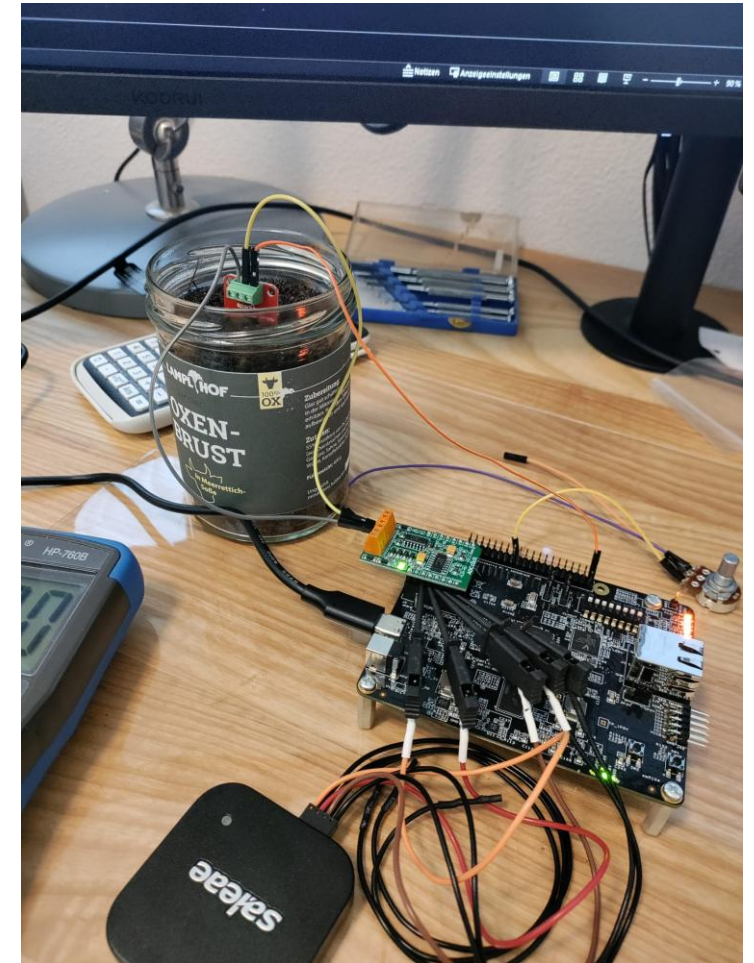
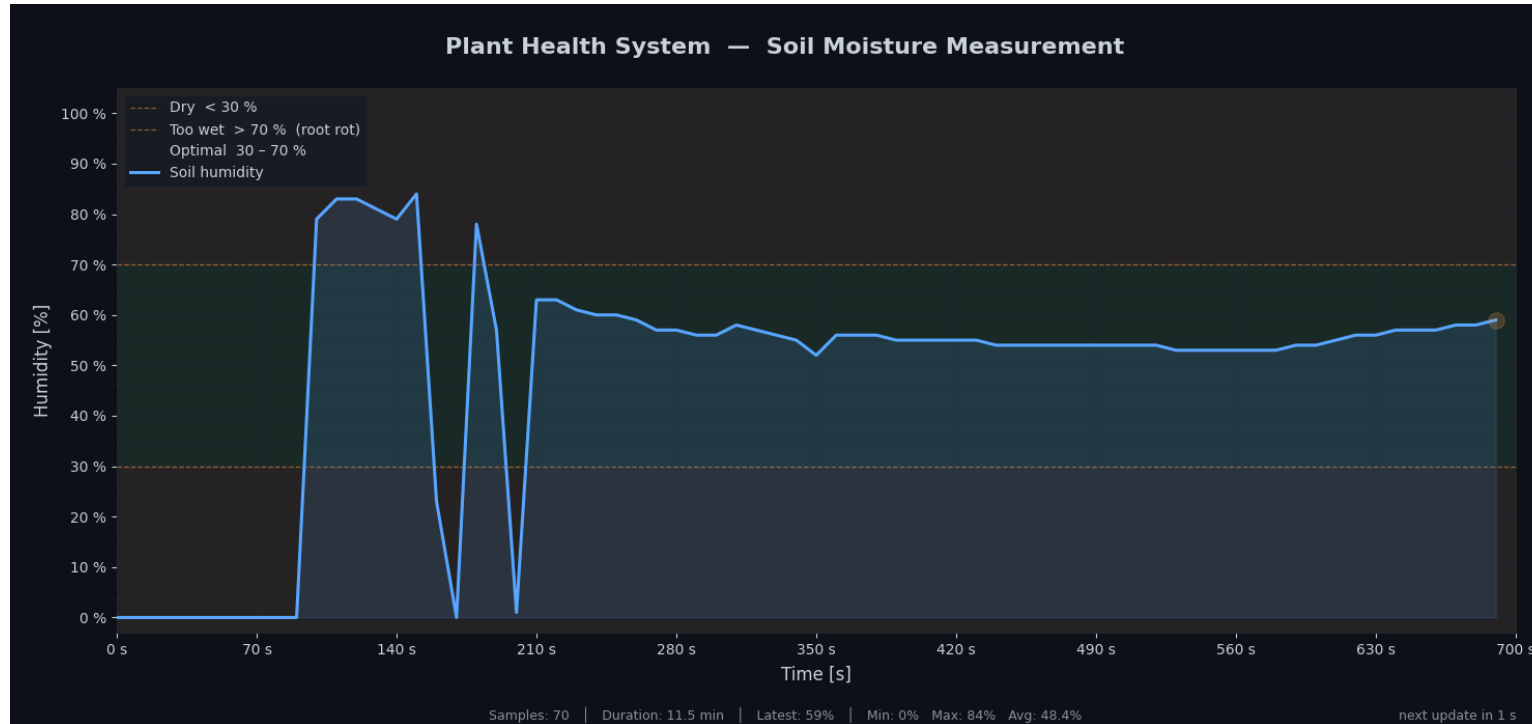
The extracted sections occur periodically (measurement command and ADC response). By measuring in the logic analyzer GUI, the 1 ms interval could be verified. ¹¹

Verification Results: Digital values generated by a potentiometer-defined voltage

Spannung zwischen CH0 und GND	Digitaler Wert
0,00V	3
0,25V	322
0,50V	642
0,75V	950
1,0V	1260
1,25V	1574
1,5V	1885
1,75V	2200
2,0V	2513
2,25V	2817
2,5V	3127
2,75V	3443
3,0V	3752
3,27V	4090



Verification Results: Python Visualization



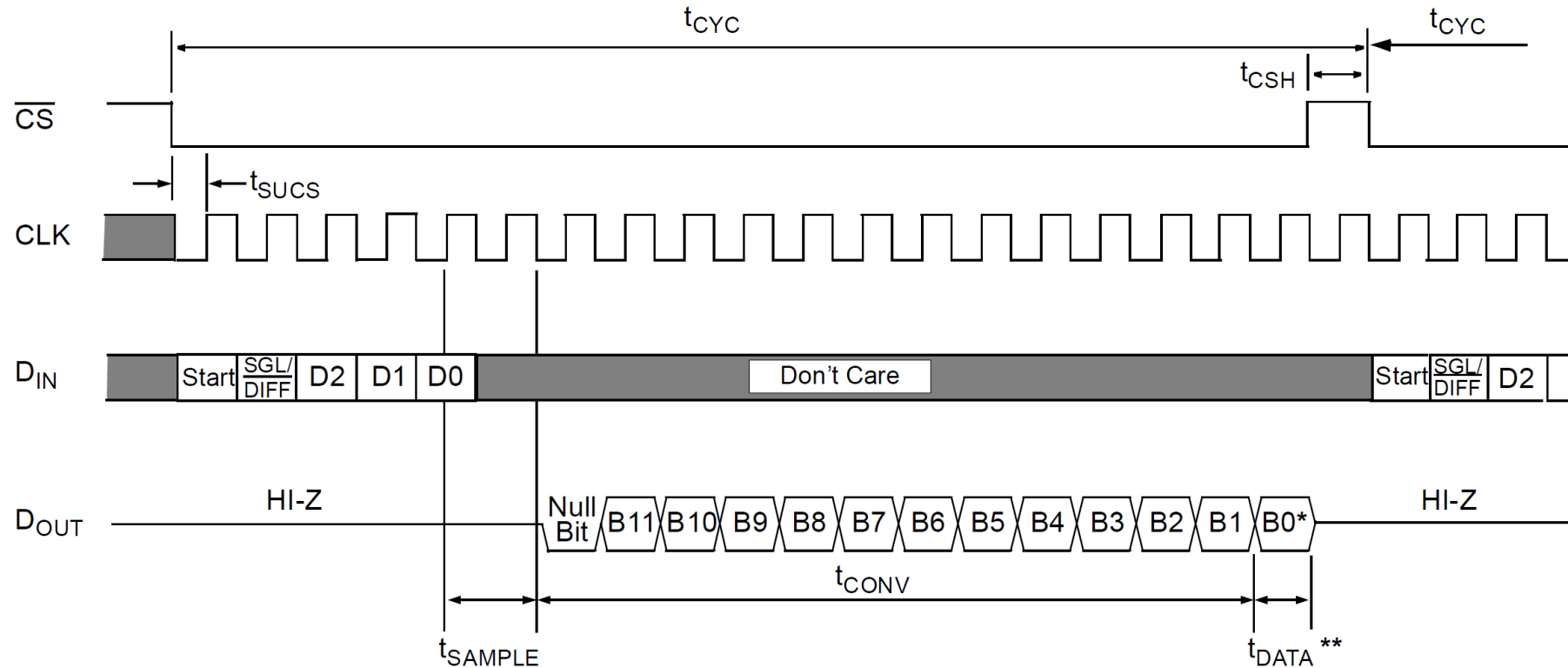
Project Closeout

- **Achievements:** As planned, one ADC channel for soil moisture measurement is functioning correctly, which has been verified against the defined requirements.
- **non-conformances:** The temperature measurement and the integration of additional sensors (there is also software work necessary!) have not yet been implemented.
- **Lessons learned:** Developing the functionality itself can be done relatively quickly; however, defining requirements and performing verification require significant time and often involve extensive discussion and alignment.

**Thank you
for your attention**



Backup-Slides: SPI Communication



* After completing the data transfer, if further clocks are applied with $\overline{CS}$ low, the A/D converter will output LSB first data, followed by zeros indefinitely (see Figure 5-2 below).

** t_{DATA} : during this time, the bias current and the comparator power down while the reference input becomes a high impedance node, leaving the CLK running to clock out the LSB-first data or zeros.